1.1.11. Knapsack... 17:07

Write a program to implement the 0/1 Knapsack Problem. Given N items, each having a value and a weight, and a knapsack with capacity W, determine the maximum total value that can be obtained by selecting a subset of the items such that the total weight does not exceed W. An item can either be included completely or excluded completely from the knapsack. Fractional selection of items is not allowed.

Input Format:

- The first line contains an integer N, representing the number of items.
- The second line contains N space-separated integers representing the values of the items.
- The third line contains N space-separated integers representing the weights of the items.
- The fourth line contains an integer W, representing the capacity of the knapsack.

Output Format:

- Print a single integer representing the maximum value that can be obtained without exceeding the knapsack capacity.

Constraints:

- $1 \leq N \leq 1000$
- $1 \leq W \leq 1000$
- $1 \leq value[i], weight[i] \leq 1000$

Sample Input:

```
3
1 2 3
```

Explorer | maxValue... | Submit | Debugger

```c
#include <stdio.h>
#include <stdlib.h>
int max(int a, int b) {
    return (a > b) ? a : b;
}
int main() {
    int N;
    if (scanf("%d", &N) != 1) return 0;
    int *values = (int *)malloc(N * sizeof(int));
    for (int i = 0; i < N; i++) {
        if (scanf("%d", &values[i]) != 1) return 0;
    }

    int *weights = (int *)malloc(N * sizeof(int));
    for (int i = 0; i < N; i++) {
        if (scanf("%d", &weights[i]) != 1) return 0;
    }

    int W;
    if (scanf("%d", &W) != 1) return 0;
    int *dp = (int *)calloc(W + 1, sizeof(int));
    for (int i = 0; i < N; i++) {
        for (int w = W; w >= weights[i]; w--) {
            dp[w] =
```

‹ Prev | Reset | Submit | Next ›

```
4 5 1
4
```

Sample Output:

```
3
```

Explanation: The knapsack capacity is 4. Among the given items, selecting the third item (value = 3, weight = 1) yields the maximum value without exceeding the capacity. Therefore, the output is 3.

Note:

- Each item can be selected at most once.
- Fractional selection of items is not allowed.
- The total weight of the selected items must not exceed the knapsack capacity W.

Sample Test Cases +

```
        weights[i]] +
        values[i]);
25              }
26          }
27          printf("%d\n",
        dp[W]);
28
29          free(values);
30          free(weights);
31          free(dp);
32          return 0;
33      }
34
35
```